Notes on Machine Learning

Course of Machine Learning

Master Degree in Computer Science

University of Rome "Tor Vergata"

Giorgio Gambosi

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Chapter 1

Linear regression: A Summary

Linear Models and Basis Function Expansion

- Linear models assume target prediction as linear combination of input features.
- For d-dimensional input $\mathbf{x} = (x_1, \dots, x_d)^T$, hypothesis function:

$$h(\mathbf{x}; w_1, \dots, w_d, b) = w_1 x_1 + w_2 x_2 + \dots + w_d x_d + b \tag{1.1}$$

- b is the bias, allowing non-zero prediction when all features are zero.
- In vector form: $h(\mathbf{x}; \mathbf{w}, b) = \mathbf{w}^T \mathbf{x} + b$.
- **Two types of linearity:**
  - Linear in **features** $\mathbf{x}$ (flat hyperplane in input space).
  - Linear in **parameters** $\mathbf{w}$ and b (convex objective functions).

Feature Engineering and Basis Functions

- Map original input $\mathbf{x} \in \mathbb{R}^d$ to higher-dimensional space using basis functions $\{\phi_1, \dots, \phi_m\}$:

$$\boldsymbol{\phi}(\mathbf{x}) = (\phi_1(\mathbf{x}), \dots, \phi_m(\mathbf{x}))^T \tag{1.2}$$

- Model becomes:

$$h(\mathbf{x}; \mathbf{w}, b) = \sum_{j=1}^{m} w_j \phi_j(\mathbf{x}) + b \tag{1.3}$$

- Still linear in parameters $\mathbf{w}$; only the "ingredients" change.

Common Types of Basis Functions

- **Polynomial (Global):** $\phi_j(x) = x^j$.
- **Gaussian (Local):** $\phi_j(x) = \exp\left(-\frac{(x-\mu_j)^2}{2s^2}\right)$.
- **Sigmoid (Local):** $\phi_j(x) = \sigma\left(\frac{x-\mu_j}{s}\right) = \frac{1}{1+e^{-\frac{x-\mu_j}{s}}}$.
- **Hyperbolic Tangent (Local):** $\phi_j(x) = \tanh(x) = 2\sigma(x) - 1$.

The Design Matrix

- Dataset of n observations:

$$\mathbf{X} = \begin{pmatrix} - & \mathbf{x}_1 & - \\ & \vdots & \\ - & \mathbf{x}_n & - \end{pmatrix} \tag{1.4}$$

- After applying basis functions, **Design Matrix** $\mathbf{\Phi}$:

$$\mathbf{\Phi} = \begin{pmatrix} \phi_1(\mathbf{x}_1) & \phi_2(\mathbf{x}_1) & \cdots & \phi_m(\mathbf{x}_1) \\ \phi_1(\mathbf{x}_2) & \phi_2(\mathbf{x}_2) & \cdots & \phi_m(\mathbf{x}_2) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_1(\mathbf{x}_n) & \phi_2(\mathbf{x}_n) & \cdots & \phi_m(\mathbf{x}_n) \end{pmatrix} \tag{1.5}$$

- All derivations apply equally to original $\mathbf{X}$ and transformed $\mathbf{\Phi}$.

Polynomial Regression Example:

- Training set: $\{(x_1, t_1), \dots, (x_n, t_n)\}$ with true relationship $t = \sin(2\pi x)$ corrupted by Gaussian noise $\varepsilon \sim \mathcal{N}(0, \sigma^2)$:

$$t_i = \sin(2\pi x_i) + \varepsilon_i \tag{1.6}$$

- Polynomial model of degree m:

$$h(x; \mathbf{w}, b) = \sum_{j=1}^{m} w_j x^j + b \tag{1.7}$$

- Critical choice of m: low $m \Rightarrow$ underfitting; high $m \Rightarrow$ overfitting.

Mathematical Optimization: Convexity and Global Minima

- Quadratic loss leads to convex error function $E(\overline{\mathbf{w}})$ (sum of convex functions).
- Any local minimum is **global minimum**.
- Setting gradient to zero yields linear system.
- Partial derivatives:

$$\frac{\partial E(\overline{\mathbf{w}})}{\partial w_k} = 2 \sum_{i=1}^{n} \left(\sum_{j=1}^{d} w_j x_{ij} + b - t_i \right) x_{ik} \quad k = 1, \dots, d$$

$$\frac{\partial E(\overline{\mathbf{w}})}{\partial b} = 2 \sum_{i=1}^{n} \left(\sum_{j=1}^{d} w_j x_{ij} + b - t_i \right)$$

- In matrix form with **augmented design matrix**:

$$\overline{\mathbf{X}} = \begin{pmatrix} 1 & x_{11} & \cdots & x_{1d} \\ 1 & x_{21} & \cdots & x_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \cdots & x_{nd} \end{pmatrix} \tag{1.8}$$

- **Normal equations** (assuming $\overline{\mathbf{X}}^T\overline{\mathbf{X}}$ non-singular):

$$\overline{\mathbf{w}}^* = (\overline{\mathbf{X}}^T\overline{\mathbf{X}})^{-1}\overline{\mathbf{X}}^T\mathbf{t} \tag{1.9}$$

Numerical Optimization: Gradient Descent

- Iterative method for large datasets where matrix inversion is expensive.
- Start with initial guess $\overline{\mathbf{w}}^{(0)}$.
- Update in direction of **steepest descent** (negative gradient):

$$\overline{\mathbf{w}}^{(s)} := \overline{\mathbf{w}}^{(s-1)} - \eta\nabla E(\overline{\mathbf{w}})\Big|_{\overline{\mathbf{w}}^{(s-1)}} \tag{1.10}$$

- Component-wise with learning rate η:

$$w_k^{(s)} := w_k^{(s-1)} - 2\eta\sum_{i=1}^{n} r_i(\overline{\mathbf{w}}^{(s-1)})x_{ik}$$
$$b^{(s)} := b^{(s-1)} - 2\eta\sum_{i=1}^{n} r_i(\overline{\mathbf{w}}^{(s-1)})$$

- Matrix form:

$$\overline{\mathbf{w}}^{(s)} := \overline{\mathbf{w}}^{(s-1)} - 2\eta\sum_{i=1}^{n} r_i(\overline{\mathbf{w}}^{(s-1)})\overline{\mathbf{x}}_i \tag{1.11}$$

- Update proportional to residual r_i: perfect prediction $\Rightarrow$ no update.

Model Selection for Polynomial Degree M:

- Increasing $M \Rightarrow$ more degrees of freedom, better training fit, but risk of overfitting.
- If $M + 1 = n$, model can pass exactly through every point (zero training error) but overfits.
- **Generalization evaluation:**
  1. Generate training set $\mathcal{T}_{train}$ from $t = \sin(2\pi x) + \varepsilon$.

2. Generate separate test set $\mathcal{T}_{test}$ from same process.
3. For each M, fit $\mathbf{w}^*$ on $\mathcal{T}_{train}$, evaluate RMS error on $\mathcal{T}_{test}$:

$$E_{RMS}(\mathbf{w}^*, \mathcal{T}_{test}) = \sqrt{\frac{1}{|\mathcal{T}_{test}|} \sum_{(x_i,t_i)\in\mathcal{T}_{test}} \left(\sum_{j=0}^{M} w_j^* x_i^j - t_i \right)^2} \tag{1.12}$$

- **Error trends:**
  - Training error decreases monotonically with M.
  - Test error decreases initially (capturing signal), then increases sharply (overfitting).

Regularization in Linear Models

- Overfitting depends on interplay between model complexity and data quantity.
- Larger training sets allow higher model complexity.
- When data is limited, **regularization** explicitly controls complexity.

Regularization: Controlling Complexity via the Cost Function

- Total objective:

$$E_{total}(\overline{\mathbf{w}}) = E_D(\overline{\mathbf{w}}) + \lambda E_W(\overline{\mathbf{w}}) \tag{1.13}$$

- $E_D(\overline{\mathbf{w}})$: empirical risk (data fit term).
- $E_W(\overline{\mathbf{w}})$: **regularization term** (penalizes extreme parameter values).
- λ: **regularization coefficient** controlling trade-off.
  - Small $\lambda \Rightarrow$ prioritize data fit (risk overfitting).
  - Large $\lambda \Rightarrow$ prioritize small weights (risk underfitting).

Ridge Regression: Quadratic Regularization

- For quadratic loss, common regularization:

$$E_W(\overline{\mathbf{w}}) = \frac{1}{2}\overline{\mathbf{w}}^T\overline{\mathbf{w}} = \frac{1}{2}\sum_{i=1}^{m} w_i^2 + \frac{b^2}{2} \tag{1.14}$$

- Total loss:

$$E(\mathbf{w}) = \frac{1}{2}\sum_{i=1}^{n} r_i(\mathbf{w})^2 + \frac{\lambda}{2}\mathbf{w}^T\mathbf{w} = \frac{1}{2}\boldsymbol{r}(\mathbf{w})^T\boldsymbol{r}(\mathbf{w}) + \frac{\lambda}{2}\mathbf{w}^T\mathbf{w} \tag{1.15}$$

- Residual vector: $\boldsymbol{r}(\overline{\mathbf{w}}) = \overline{\mathbf{X}}\overline{\mathbf{w}} - \mathbf{t}$.
- **Ridge regression** (Tikhonov regularization) closed-form solution:

$$\overline{\mathbf{w}} = (\lambda \mathbf{I} + \overline{\mathbf{X}}^T \overline{\mathbf{X}})^{-1} \overline{\mathbf{X}}^T \mathbf{t} \tag{1.16}$$

- $\lambda \mathbf{I}$ strengthens diagonal, ensures well-conditioned matrix even with correlated features.

Generalized Regularization and Lasso

- General form with power q:

$$E(\overline{\mathbf{w}}) = \frac{1}{2} \sum_{i=1}^{n} r_i(\overline{\mathbf{w}})^2 + \frac{\lambda}{2} \sum_{j=1}^{d} |w_j|^q + \frac{|b|^q}{2} \tag{1.17}$$

- $q = 1$: **Lasso** (Least Absolute Shrinkage and Selection Operator).
- Lasso favors **sparse models** (many weights exactly zero).
- Performs automatic feature selection.

Geometric Comparison: Ridge vs. Lasso Level Curves

- Minimization equivalent to constrained optimization: minimize $E_D(\overline{\mathbf{w}})$ subject to $E_W(\overline{\mathbf{w}}) \leq \eta$, with $\eta \propto 1/\lambda$.
- **Ridge ($q = 2$):** Constraint $\sum w_j^2 \leq \eta$ defines a **hypersphere** (circle in 2D). Smooth boundary $\Rightarrow$ weights shrink toward zero but rarely become exactly zero.
- **Lasso ($q = 1$):** Constraint $\sum |w_j| \leq \eta$ defines a **hyperdiamond** (diamond in 2D). Sharp corners on axes $\Rightarrow$ elliptical error contours likely hit corners, setting other weights to exactly zero.
- Lasso's "vertex-hitting" property produces sparse models.

Ridge Regularization in Polynomial Regression:

- Objective:

$$E(\overline{\mathbf{w}}) = \frac{1}{2} \sum_{i=1}^{n} r_i(\overline{\mathbf{w}})^2 + \frac{\lambda}{2} ||\overline{\mathbf{w}}||^2 \tag{1.18}$$

- Penalty term shrinks coefficients, preventing high-frequency oscillations.
- **Effect of λ:**
  - Small λ: negligible penalty $\Rightarrow$ low training error, high test error (overfitting).
  - Large λ: penalty dominates $\Rightarrow$ oversmoothed, high error on both sets (underfitting).
  - Intermediate λ: balanced $\Rightarrow$ minimal test error (best generalization).

Multi-Dataset Analysis (Bias-Variance):

- $L = 100$ training sets, each $n = 25$, target $f(x) = \sin(2\pi x)$, $m = 24$ Gaussian basis functions (prone to overfitting).
- For each λ, fit $h^{(i)}(x)$ on each dataset.
- **High λ (high regularization):** Low variance (models similar), high bias (average poor fit).
- **Low λ (low regularization):** High variance (models very different), low bias (average close to true function).
- **Intermediate λ:** Optimal bias-variance trade-off.
- Total error = bias^2 + variance; optimal λ at minimum of this sum.

Probabilistic model for regression

- Assume joint distribution $p(\mathbf{x}, t) = p_M(\mathbf{x})p_C(t|\mathbf{x})$, with $p_M(\mathbf{x})$ uniform.
- Observed target t normally distributed around prediction $h(\mathbf{x}; \overline{\mathbf{w}})$ with variance σ^2 (precision $\beta = 1/\sigma^2$):

$$p(t|\mathbf{x}; \overline{\mathbf{w}}, \beta) = \mathcal{N}(t|h(\mathbf{x}; \overline{\mathbf{w}}), \beta^{-1}) \tag{1.19}$$

- $h(\mathbf{x}; \overline{\mathbf{w}})$ is the mean, β is certainty (higher $\beta \Rightarrow$ tighter clustering).

Maximum Likelihood Estimation (MLE)

- Likelihood (i.i.d. observations):

$$L(\overline{\mathbf{w}}, \beta|\mathbf{X}, \mathbf{t}) = \prod_{i=1}^{n} \mathcal{N}(t_i|h(\mathbf{x}_i; \overline{\mathbf{w}}), \beta^{-1}) \tag{1.20}$$

- Log-likelihood:

$$l(\overline{\mathbf{w}}, \beta|\mathbf{X}, \mathbf{t}) = -\frac{\beta}{2} \sum_{i=1}^{n} r_i(\overline{\mathbf{w}})^2 + \frac{n}{2} \log \beta + c$$

- Maximizing wrt $\overline{\mathbf{w}} \equiv$ minimizing sum of squared errors (Least Squares).
- Maximizing wrt β:

$$\beta_{ML}^{-1} = \frac{1}{n} \sum_{i=1}^{n} r_i(\overline{\mathbf{w}})^2 \tag{1.21}$$

- Predictive distribution (point estimate):

$$p(t|\mathbf{x}; \overline{\mathbf{w}}_{ML}, \beta_{ML}) = \mathcal{N}(t|h(\mathbf{x}; \overline{\mathbf{w}}_{ML}), \beta_{ML}^{-1}) \tag{1.22}$$

The Bayesian Approach and Conjugate Priors

- Treat parameters $\overline{\mathbf{w}}$ as random variables with **prior distribution**.
- Zero-mean Gaussian prior with precision α:

$$p(\overline{\mathbf{w}}|\alpha) = \mathcal{N}(\overline{\mathbf{w}}; \mathbf{0}, \alpha^{-1}\mathbf{I}) = \left(\frac{\alpha}{2\pi}\right)^{\frac{m}{2}} e^{-\frac{\alpha}{2}\overline{\mathbf{w}}^T\overline{\mathbf{w}}} \tag{1.23}$$

- Gaussian is **conjugate prior** for mean of Gaussian likelihood $\Rightarrow$ posterior also Gaussian.
- For general prior $p(\overline{\mathbf{w}}) = \mathcal{N}(\overline{\mathbf{w}}; \mathbf{m}_0, \Sigma_0)$, posterior:

$$\Sigma_p = (\Sigma_0^{-1} + \beta\overline{\mathbf{X}}^T\overline{\mathbf{X}})^{-1}$$
$$\mathbf{m}_p = \Sigma_p(\Sigma_0^{-1}\mathbf{m}_0 + \beta\overline{\mathbf{X}}^T\mathbf{t})$$

- For zero-mean prior ($\mathbf{m}_0 = \mathbf{0}$, $\Sigma_0^{-1} = \alpha\mathbf{I}$):

$$\Sigma_p = (\alpha\mathbf{I} + \beta\overline{\mathbf{X}}^T\overline{\mathbf{X}})^{-1}$$
$$\mathbf{m}_p = \beta\Sigma_p\overline{\mathbf{X}}^T\mathbf{t}$$

- Posterior mean $\mathbf{m}_p$ = Ridge Regression solution.
- As $\alpha \to 0$, $\mathbf{m}_p \to \overline{\mathbf{w}}_{ML}$.

Sequential Bayesian Learning

- Posterior after observing dataset can serve as prior for next dataset.
- For independent training sets $\mathcal{T}_1, \dots, \mathcal{T}_n$:

$$p(\overline{\mathbf{w}}|\mathcal{T}_1) \propto p(\mathcal{T}_1|\overline{\mathbf{w}})p(\overline{\mathbf{w}})$$
$$p(\overline{\mathbf{w}}|\mathcal{T}_1, \dots, \mathcal{T}_k) \propto p(\mathcal{T}_k|\overline{\mathbf{w}})p(\overline{\mathbf{w}}|\mathcal{T}_1, \dots, \mathcal{T}_{k-1})$$

- Enables online learning from data streams.
- Posterior covariance update:

$$\Sigma_{n+1}^{-1} = \Sigma_n^{-1} + \beta\overline{\mathbf{x}}_{n+1}\overline{\mathbf{x}}_{n+1}^T \tag{1.24}$$

- Each observation adds positive semi-definite term $\Rightarrow$ precision increases, covariance decreases.
- Using Sherman-Morrison formula:

$$\Sigma_{n+1} = \Sigma_n - \frac{\Sigma_n\overline{\mathbf{X}}\overline{\mathbf{X}}^T\Sigma_n}{\beta^{-1} + \overline{\mathbf{X}}^T\Sigma_n\overline{\mathbf{X}}} \tag{1.25}$$

- $\Delta\Sigma = \Sigma_n - \Sigma_{n+1}$ positive semi-definite $\Rightarrow$ uncertainty ellipsoid only shrinks.

Sequential Bayesian Learning for Simple Linear Regression:

- Model: $h(x; b, w_1) = w_1 x + b$, true parameters $a_0 = -0.3$, $a_1 = 0.5$, noise $\sigma = 0.2$.
- Prior: bivariate Gaussian centered at origin, $\mathbf{\Sigma} = 0.04\mathbf{I}$.
- **After 1 point:** Posterior elongated into stripe; sampled lines pass near first observation.
- **After 2 points:** Posterior concentrated at intersection of constraints; lines pass near both points.
- **After many points:** Posterior highly concentrated around true parameters; sampled lines nearly indistinguishable.
- Visual evolution shows transition from high uncertainty (broad prior) to high certainty (sharp posterior).

Maximum a Posteriori (MAP)

- MAP estimate = mode of posterior distribution.
- Log-posterior (ignoring evidence constant):

$$\log p(\overline{\mathbf{w}}|\mathbf{X}, \mathbf{t}; \alpha, \beta) = \log p(\mathbf{t}|\mathbf{X}, \overline{\mathbf{w}}; \beta) + \log p(\overline{\mathbf{w}}; \alpha) + \text{const} \tag{1.26}$$

- Maximization equivalent to minimization:

$$\overline{\mathbf{w}}_{MAP} = \underset{\overline{\mathbf{w}}}{\operatorname{argmin}} \left(-\log p(\mathbf{t}|\mathbf{X}, \overline{\mathbf{w}}; \beta) - \log p(\overline{\mathbf{w}}; \alpha)\right) \tag{1.27}$$

- For Gaussian likelihood and prior:

$$\log p(\mathbf{t}|\mathbf{X}, \overline{\mathbf{w}}; \beta) = \frac{n}{2} \log \frac{\beta}{2\pi} - \frac{\beta}{2} \sum_{i=1}^{n} r_i(\overline{\mathbf{w}})^2$$

$$\log p(\overline{\mathbf{w}}; \alpha) = \frac{d+1}{2} \log \frac{\alpha}{2\pi} - \frac{\alpha}{2} \sum_{j=0}^{d} \overline{w}_j^2$$

- Dropping constants, maximizing posterior $\equiv$ minimizing:

$$\frac{\beta}{2} \sum_{i=1}^{n} r_i(\overline{\mathbf{w}})^2 + \frac{\alpha}{2} \sum_{j=0}^{d} \overline{w}_j^2 \propto \frac{1}{2} \sum_{i=1}^{n} r_i(\mathbf{w})^2 + \frac{\alpha}{2\beta} \sum_{j=0}^{d} \overline{w}_j^2 \tag{1.28}$$

- This is **Ridge Regression** with $\lambda = \frac{\alpha}{\beta}$.
- Regularization = consequence of Gaussian prior on parameters.
- Predictive distribution with MAP:

$$p(t|\mathbf{x}, \overline{\mathbf{w}}_{MAP}; \beta_{MAP}) = \mathcal{N}(t|h(\mathbf{x}; \overline{\mathbf{w}}_{MAP}), \beta_{MAP}^{-1}) \tag{1.29}$$

- β_{MAP} equals β_{ML} (noise variance estimate).

Approaches to prediction in linear regression

Classical (Frequentist)

- Point estimate $\overline{\mathbf{w}}_{LS}$ by minimizing quadratic cost (or maximizing likelihood under Gaussian noise).
- Regularization can be added.
- Prediction: $t = \overline{\mathbf{x}}^T \overline{\mathbf{w}}_{LS} = \mathbf{x}^T \mathbf{w}_{LS} + b_{LS}$.

Bayesian point estimation (MAP)

- Derive posterior $p(\overline{\mathbf{w}}|\mathbf{X}, \mathbf{t}; \alpha, \beta)$, compute mode $\overline{\mathbf{w}}_{MAP}$.
- Equivalent to classical with $\lambda = \alpha/\beta$.
- Prediction: Gaussian $\mathcal{N}(t|\overline{\mathbf{x}}^T \overline{\mathbf{w}}_{MAP}, \beta)$ with mean $\overline{\mathbf{x}}^T \overline{\mathbf{w}}_{MAP}$ and variance β^{-1}.
- Distribution built from assumed noise, not directly from posterior.

Fully Bayesian

- Derive **predictive distribution** $p(t|\mathbf{x}, \mathbf{X}, \mathbf{t})$ by marginalizing over $\overline{\mathbf{w}}$:

$$p(t|\mathbf{x}, \mathbf{t}, \mathbf{X}; \alpha, \beta) = \int p(t|\mathbf{x}, \overline{\mathbf{w}}; \beta) p(\overline{\mathbf{w}}|\mathbf{t}, \mathbf{X}; \alpha, \beta) d\overline{\mathbf{w}} \tag{1.30}$$

- Convolution of two Gaussians yields Gaussian predictive distribution:

$$p(t|\mathbf{x}, \mathbf{X}, \mathbf{t}; \alpha, \beta) = \mathcal{N}(t; m(\mathbf{x}), \sigma^2(\mathbf{x})) \tag{1.31}$$

- Mean:

$$m(\mathbf{x}) = \beta \overline{\mathbf{x}}^T \Sigma \overline{\mathbf{X}}^T \mathbf{t} \tag{1.32}$$

- Variance:

$$\sigma^2(\mathbf{x}) = \underbrace{\frac{1}{\beta}}_{\text{aleatoric (noise)}} + \underbrace{\overline{\mathbf{x}}^T \Sigma \overline{\mathbf{x}}}_{\text{epistemic (parameter uncertainty)}} \tag{1.33}$$

- $\Sigma = (\alpha \mathbf{I} + \beta \overline{\mathbf{X}}^T \overline{\mathbf{X}})^{-1}$.
- Epistemic term shrinks near training points (where parameters well-constrained).
- Variance larger in "data-void" regions.

Evolution of Bayesian Predictive Distribution:

- Model: $m = 9$ Gaussian basis functions (complex, prone to overfitting in frequentist setting).
- Training sets of size $n = 1, 2, 4, 25$ from $f(x) = \sin(2\pi x)$ with noise.

- $n = 1$: Mean forced through single point, nearly flat elsewhere; variance very wide; sampled curves wildly different.
- $n = 2$: Mean connects two points; variance still large in gaps; sampled curves more aligned but still diverse.
- $n = 4$: S-shape emerges; variance "pinches" at training points, widens between them; sampled curves consistent in oscillation.
- $n = 25$: Mean excellent approximation; variance collapsed and nearly uniform; sampled curves almost indistinguishable.
- Bayesian framework provides "margin of error" that naturally shrinks with more data.

Fully Bayesian Regression and Hyperparameter Marginalization

- Ideally marginalize over hyperparameters α, β with hyperpriors:

$$p(t|\mathbf{x}, \mathbf{X}, \mathbf{t}) = \int p(t|\mathbf{x}, \overline{\mathbf{w}}; \beta) p(\overline{\mathbf{w}}|\mathbf{X}, \mathbf{t}; \alpha, \beta) p(\alpha, \beta|\mathbf{X}, \mathbf{t})\, d\overline{\mathbf{w}}\, d\alpha\, d\beta \tag{1.34}$$

- Non-linear dependence of weight posterior on α, β makes integral analytically intractable.
- Approximation methods needed.

Approximation Methods

Laplace Approximation: Find mode of hyperparameter posterior, second-order Taylor expansion around mode $\Rightarrow$ Gaussian approximation.

$$\ln f(\mathbf{z}) \approx \ln f(\mathbf{z}_0) - \frac{1}{2}(\mathbf{z} - \mathbf{z}_0)^T \mathbf{A}(\mathbf{z} - \mathbf{z}_0) \tag{1.35}$$

where $\mathbf{A}$ is Hessian of negative log-probability.

Variational Inference (VI): Define simpler family $q(\overline{\mathbf{w}}, \alpha, \beta)$ (often with mean-field independence assumption), minimize KL divergence between q and true posterior.

Markov Chain Monte Carlo (MCMC): Draw samples $\{\overline{\mathbf{w}}^{(l)}, \alpha^{(l)}, \beta^{(l)}\}_{l=1}^{L}$ from posterior via Markov chain. Approximate predictive:

$$p(t|\mathbf{x}, \mathbf{X}, \mathbf{t}) \approx \frac{1}{L} \sum_{l=1}^{L} p(t|\mathbf{x}, \overline{\mathbf{w}}^{(l)}, \beta^{(l)}) \tag{1.36}$$

As $L \to \infty$, converges to true integral.

Choice depends on dataset size and required precision.

The Evidence Framework (Empirical Bayes)

- Treat α, β as quantities to be inferred from data by maximizing **marginal likelihood** (evidence):

$$p(\mathbf{t}|\alpha, \beta) = \int p(\mathbf{t}|\overline{\mathbf{w}}, \beta)\, p(\overline{\mathbf{w}}|\alpha)\, d\overline{\mathbf{w}} \tag{1.37}$$

- Averages over all possible $\overline{\mathbf{w}}$ weighted by prior $\Rightarrow$ evaluates model class, not just one fitted model.
- For Gaussian likelihood and prior, integral analytic. Log-evidence:

$$\log p(\mathbf{t}|\alpha, \beta) = \frac{m}{2}\log\alpha + \frac{n}{2}\log\beta - E(\mathbf{m}_p) - \frac{1}{2}\log|\mathbf{A}| - \frac{n}{2}\log(2\pi) \tag{1.38}$$

  where:

  - $\mathbf{A} = \Sigma^{-1} = \alpha\mathbf{I} + \beta\overline{\mathbf{X}}^T\overline{\mathbf{X}}$ (Hessian, inverse posterior covariance).
  - $E(\mathbf{m}_p) = \frac{\beta}{2}\|\mathbf{t} - \overline{\mathbf{X}}\mathbf{m}_p\|^2 + \frac{\alpha}{2}\mathbf{m}_p^T\mathbf{m}_p$ (regularized error at posterior mean).
- First two terms favor large α, β; remaining terms penalize complexity/poor fit.
- Maximizing log-evidence automatically balances data fit and model complexity.

Effective Number of Parameters γ

- Let η_i be eigenvalues of $\beta\overline{\mathbf{X}}^T\overline{\mathbf{X}}$.
- For each direction i, ratio $\lambda_i = \frac{\eta_i}{\alpha+\eta_i}$:
  - $\eta_i \gg \alpha$: data dominates, $\lambda_i \approx 1$ (parameter determined by data).
  - $\eta_i \ll \alpha$: prior dominates, $\lambda_i \approx 0$ (parameter shrunk toward zero).
- Effective number of parameters:

$$\gamma = \sum_{i=1}^{m} \frac{\eta_i}{\alpha + \eta_i} \tag{1.39}$$

- $\gamma \approx m \Rightarrow$ data determines nearly all parameters.
- $\gamma \ll m \Rightarrow$ most parameters constrained by prior (effective degrees of freedom $\ll m$).
- Spring analogy: each weight pulled by prior spring (stiffness α) and data spring (stiffness η_i). γ counts weights significantly pulled by data.

Estimation of Optimal α and β

- Setting derivatives of log-evidence to zero gives re-estimation formulas:

$$\alpha = \frac{\gamma}{\mathbf{m}_p^T\mathbf{m}_p}$$
$$\frac{1}{\beta} = \frac{1}{n-\gamma}\sum_{i=1}^{n}(t_i - \mathbf{m}_p^T\overline{\mathbf{x}}_i)^2$$

- Large posterior weights $\Rightarrow$ smaller α (weaker regularization).
- Small posterior weights $\Rightarrow$ larger α (stronger regularization).
- β estimate corrected by effective degrees of freedom $n - \gamma$.
- $\mathbf{m}_p$ and γ depend on $\alpha, \beta \Rightarrow$ solve iteratively:
  1. Initialize α, β.

2. Compute $\mathbf{m}_p$ and γ.
3. Update α, β using formulas.
4. Repeat until convergence.

- Converges quickly, provides automated complexity control.
- If data noisy (small β), evidence maximized by increasing α (shrinkage).
- If data clean, $\gamma \to m$, model uses full expressive power.
- Relies only on training data $\Rightarrow$ no validation set or cross-validation needed.